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Clarke Transformation Module Documentation

1. Clarke Transformation Module (Clarke.h / Clarke.c)

***Disclaimer:** The source code detailed in this section was extracted from the repository maintained by the PECLAB [1].*

1.1. Overview

This module implements the **Clarke transformation** and its inverse. The transformation maps instantaneous three-phase quantities—voltages or currents in the ABC natural reference frame—into an equivalent orthogonal two-axis stationary reference frame (α - β -0), thereby decoupling the three interdependent phase variables into two independent components and a zero-sequence term. This simplification is essential for the subsequent application of vector control strategies.

1.2. Mathematical Convention: Amplitude-Invariant Clarke Transform

The implementation adopts the **amplitude-invariant** form of the Clarke transformation. Under this convention, the transformation matrix is:

$$\mathbf{T}_C = \frac{1}{3} \begin{bmatrix} 2 & -1 & -1 \\ 0 & \sqrt{3} & -\sqrt{3} \\ 1 & 1 & 1 \end{bmatrix} \quad (1)$$

which yields:

$$\alpha = \frac{1}{3} (2A - B - C) \quad (2)$$

$$\beta = \frac{1}{\sqrt{3}} (B - C) \quad (3)$$

$$0 = \frac{1}{3} (A + B + C) \quad (4)$$

Under this convention, the magnitude of α and β equals the magnitude of the original phase quantities. This is the form preferred in power-electronics control implementations where voltage and current amplitudes must be preserved for modulation purposes.

1.3. Data Structures

Two C structures are defined in `Clarke.h`:

- `FORM_ABC`: Stores the instantaneous values of a three-phase signal (A, B, C) , each declared as `double`.
- `FORM_CLARKE`: Stores the transformed components $(\alpha, \beta, 0)$, also as `double` fields named `alpha`, `beta`, and `zero`.

1.4. Implemented Functions

1.4.1. Transform_ABC_to_Clarke

```
void Transform_ABC_to_Clarke(FORM_CLARKE *clarke, float A, float B, float C);
```

Computes equations (2)–(4) and writes the results into the structure pointed to by `clarke`. The computation is fully combinational (no internal state).

1.4.2. Transform_Clarke_to_ABC

```
void Transform_Clarke_to_ABC(FORM_ABC *abc, float alpha, float beta, float zero);
```

Implements the inverse Clarke transformation, recovering the three-phase representation from the α - β -0 components:

$$A = \alpha + 0 \tag{5}$$

$$B = -\frac{1}{2}\alpha + \frac{\sqrt{3}}{2}\beta + 0 \tag{6}$$

$$C = -\frac{1}{2}\alpha - \frac{\sqrt{3}}{2}\beta + 0 \tag{7}$$

Referencias

- [1] PECLAB. (2026). *PECLAB Algorithm Codes Repository Collection*. GitHub. Retrieved from <https://github.com/orgs/PECLAB-Algotihm-Codes/repositories>